Modus ponens is an admissible closure of derivability

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub prop:transitivity (propositional derivability section).

Editorial note. The label prop:mp names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Transitivity of derivability. Same unit, \ollabel{prop:transitivity} (propositional-logic derivability section). Final step: lifts the two-premise derivation into Gamma.

Source reference. Original proposition prop-transitivity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 843–843. Prerequisite pointer invoked at proof line(s) 636; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If* $\Gamma \vdash A$ *and* $\Gamma \vdash A \to B$*, then* $\Gamma \vdash B$*.*

Proof. We have that $\{A, A \to B\} \vdash B$:

1.	A	Hyp.
2.	$A \to B$	Hyp.
3.	B	1, 2, MP

By prop-transitivity, $\Gamma \vdash B$. □